

Monsoon, 2024

B.Tech 1st Semester Examination (Special)

CHEMISTRY

Course Code : BCH23111T

Full Marks – 50

Time – 2 hours

The figure in the margin indicates full marks for the questions.

SECTION 1

1. Answer ALL questions:

(10 x 1 = 10)

(i) Which of the following is spin active nuclei?

- ☒ (a) ^{13}C
- (b) ^{12}C
- (c) ^{16}O
- (d) ^{32}S

(ii) In which of the following, $n \rightarrow \pi^*$ transition can take place

- ☒ (a) $\text{CH}_3\text{CH}=\text{CH}_2$
- (b) CH_4
- (c) CH_3COCH_3
- (d) None of the above

(iii) A good fuel should have

- (a) High ignition temperature
- (b) Low ignition temperature
- ☒ (c) Moderate ignition temperature
- (d) High moisture content

(iv) Calorie is the amount of heat required to raise the temperature of one gram of water through

- (a) One degree centigrade
- (b) One degree Fahrenheit
- (c) 2 K
- (d) None of the above

(v) Enthalpy, H is an

- ☒ (a) extensive property
- (b) intensive property
- (c) both intensive and extensive property
- (d) none of the above properties

(vi) A process occurred spontaneously. The entropy of the universe

- (a) decreased
- ☒ (b) increased

- (c) remained same
(d) all of these
- (vii) The lowest temperature at which an oil lubricant gives off vapours that ignites for a moment is called
(a) fire point
(b) flash point
(c) refractory point
(d) None of the above
- (viii) Which of the following is not a green solvent?
(a) water
(b) benzene
(c) supercritical CO₂
(d) ionic liquid
- (ix) Which of the following quantities is proportional to the probability density at a point?
(a) the wave function
(b) the square of the wave function
(c) the de Broglie wavelength
(d) the reciprocal of the de Broglie wavelength
- (x) During electrochemical corrosion the more active metal acts as
(a) Anode
(b) Cathode
(c) Anode as well as cathode
(d) None of these

SECTION 2

Answer any one question from this section (Maximum 300 words for each) (10)

2. (a) Analyse the physical significances of Entropy, Gibbs free energy and Helmholtz free energy along with mathematical expressions. (10)

OR

- (b) (i) Explain dry and wet corrosion. (3)
(ii) Discuss the various factors influencing the rate of corrosion. (4)
(iii) Write notes on Pitting corrosion. (3)

SECTION 3

Answer any two questions from this section (Maximum 300 words for each)

(10)

3. (a) Explain the different types of electronic transitions that can happen in a molecule when it absorbs UV-Visible radiation with examples.

(5)

(b) Distinguish between gross calorific value and net calorific value of a fuel.

(5)

(c) Write the Schrodinger wave equation and explain the each term involved.

(5)

(d) Discuss about setting and hardening of cement.

(5)

SECTION 4

Answer any one question from this section (Maximum 300 words for each)

(10)

4 (a) Discuss different types of Refractory materials. Discuss the characteristics of a good Lubricant.

(6+4)

OR

(b) Explain the 12 principles of Green Chemistry with examples of each principle.

(10)

SECTION 5

Answer any one question from this section (Maximum 300 words for each)

(10)

5 (a) Calculate the work performed by a system during reversible isothermal expansion of 2 moles of an ideal gas from 2 litres to 10 litres at 20 °C. (ii) Calculate the entropy change for the fusion of 1 mole of a solid which melts at 300K. The latent heat of fusion is 2.51 kJ/mol.

(6+4=10)

OR

(b) (i) Determine the gross and net calorific values of a coal sample containing carbon=80%, Hydrogen=10%, Sulphur = 2% nitrogen=3% , Oxygen=1% and ash=4%. latent heat of steam is 587 cal/g. (5)

(ii) An electron in a one-dimensional box of width 1 Å undergoes a transition from the ground state ($n=1$) to the first excited state ($n=2$). Calculate the transition energy.

(5)